Physical Chemistry Cume April 7 2023

This cumulative exam is based on the following announced topic/paper:

Maria Vanessa Balois-Oguchi, Norihiko Hayazawa, Satoshi Yasuda, Katsuyoshi Ikeda, Tien Quang Nguyen, Mary Clare Escaño, and Takuo Tanaka. Probing Strain and Doping along a Graphene Wrinkle Using Tip-Enhanced Raman Spectroscopy. J. Phys. Chem. C. 2023, **127**, 12, 5982–5990

1) 50 points

a) When graphene is grown on a substrate via chemical vapor deposition, why do wrinkles form? What are the consequences (both positive and negative) of these wrinkles on the properties of graphene?

b) Why can't nanometer sized wrinkles be measured using conventional Raman Spectroscopy? How is this overcome using Tip Enhanced Raman Spectroscopy?

c) describe the general set-up of a tip enhanced Raman experiment (feel free to use sketches). Describe the role of both the gold substrate and the gold tip, and how the distance between them determines the ability for the Raman signal to be enhanced.

2) 50 points

Figures 3 and 4 from the manuscript are given on the next page for reference:

For additional reference:

In the Raman Signal, the D band is observed at ~1330 cm^{-1}, the G band is observed at ~1600 cm^{-1} and the 2D band is observed at 2650 cm^{-1}.

It was described in the manuscript that the 2D band is more sensitive to strain than doping, while the G band is sensitive to both strain and doping.

Summarize the data that led the authors to make the following conclusions:

a) anisotropic strain is present along the wrinkle.

b) there is a position dependence for the carrier density.

c) wrinkles can affect the electronic properties of graphene

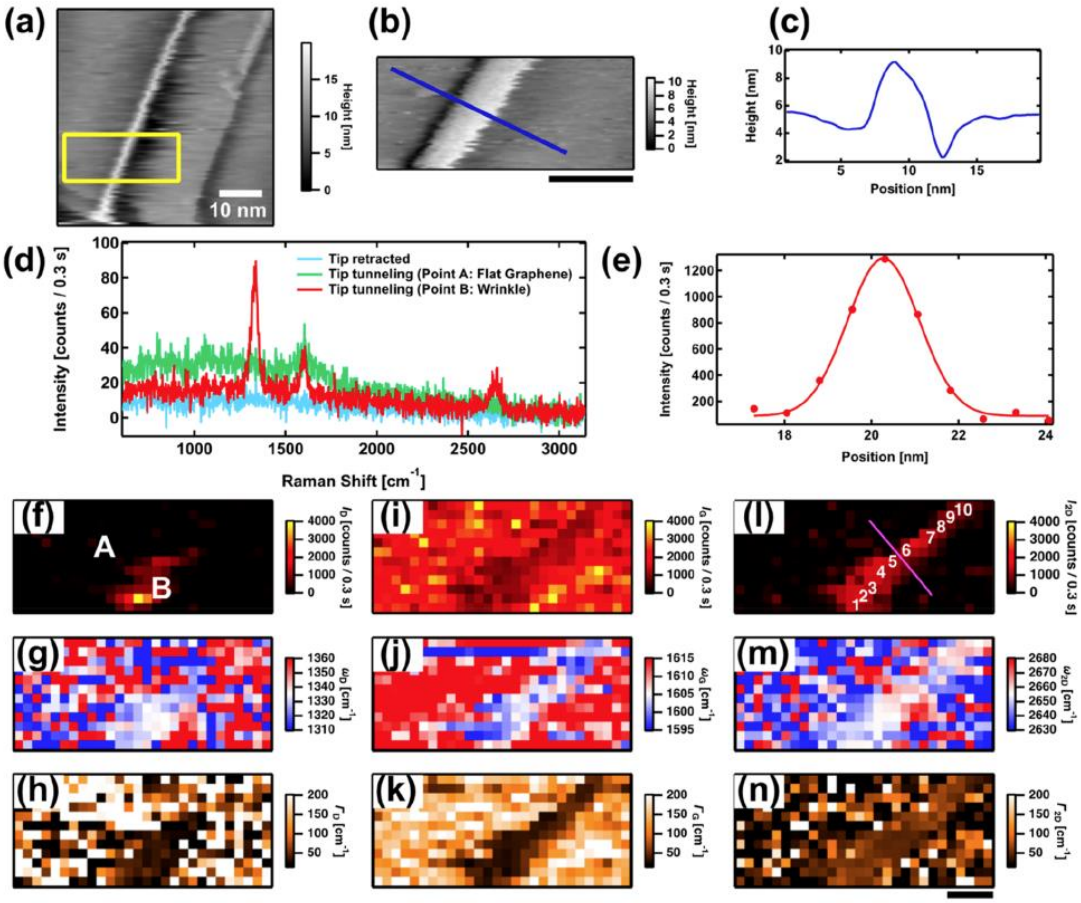

Figure 3. STM and TERS imaging of wrinkle area. (a) 50 nm × 50 nm STM topography area scan on the graphene on Au(111) sample. Yellow boxed area indicates TERS scan area (12 nm × 28 nm). (b) STM topography image of TERS scan area. The blue line indicates the location of the line profile in panel c. Scale bar represents 10 nm. (c) Line profile graphene wrinkle taken at the blue line in panel b. (d) Various spectra across the graphene sample: far-field Raman when the tip is retracted (cyan, solid), TERS spectrum when tip is tunneling at Point A (green, solid), which is flat graphene, and Point B (red, solid), which is on the wrinkle. (e) Line profile of magenta line in panel l, with a fwhm of 1.9 nm. TERS maps showing (f) intensity (I_D), (g) Raman shift (ω_D), (h) spectral width (Γ_D) of the D band; (i) intensity (I_G), (j) Raman shift (ω_G), (k) spectral width (Γ_G) of the G band; and (l) intensity (I_{2D}), (m) Raman shift (ω_{2D}), (n) spectral width (Γ_{2D}) of the 2D band. A step size of 1 nm/pixel was used for all TERS maps. Numbers 1 to 10 in panel l are the positions used for the position dependence analysis of carrier density in Figure 4d. Scale bar corresponds to 5 nm.

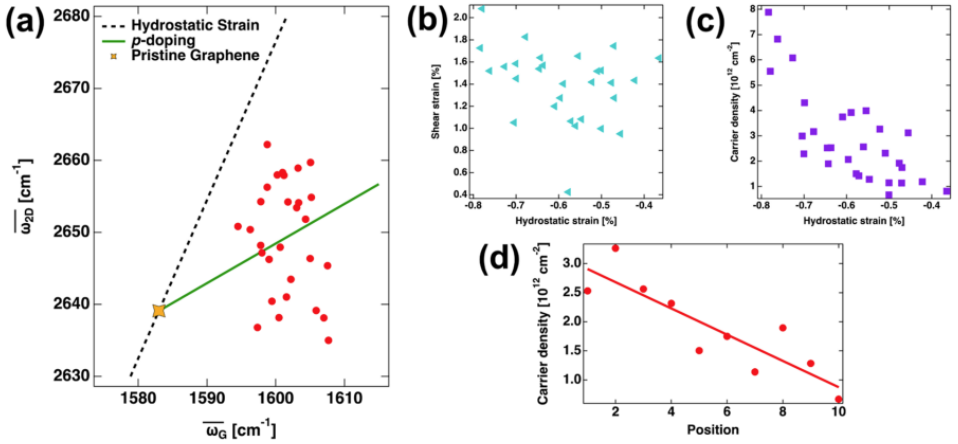

Figure 4. Strain and doping analysis. (a) Correlation plot of $\overline{\omega_{2D}}$ and $\overline{\omega_G}$. Yellow star represents the 2D and G band pair for undoped and unstrained graphene on a SiO_2/Si substrate using 632.8 nm excitation.[38,39] The expected 2D and G positions for hydrostatic strain are represented by the black dashed line, while the green solid line is for pure p-doping. (b) Plot comparing shear strain and hydrostatic strain. (c) Plot comparing carrier density and hydrostatic strain. (d) Position dependence of carrier density along the wrinkle. The positions are labeled 1 to 10 in Figure 3l.